

Problem

Find the Laplace transform of $5 \sin(at) \cos(bt)$. (*Hint: Using the exponential representation for both functions may make this problem easier.*)

Step-by-step solution

Step 1 of 5

Consider the following signal:

$$f(t) = 5 \sin(at) \cos(bt)$$

It is known that,

$$\sin \alpha \cos \beta = \frac{1}{2} (\sin(\alpha + \beta) + \sin(\alpha - \beta))$$

Therefore,

$$f(t) = 5 \left\{ \frac{1}{2} [\sin(at + bt) + \sin(at - bt)] \right\}$$

$$f(t) = \frac{5}{2} [\sin(at + bt) + \sin(at - bt)]$$

Step 2 of 5

The Laplace transform of signal $f(t)$ is,

$$L[f(t)] = \int_0^{\infty} f(t) e^{-st} dt$$

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

Substitute $\frac{5}{2} [\sin(at + bt) + \sin(at - bt)]$ for $f(t)$ in the equation.

$$\begin{aligned} F(s) &= \int_0^{\infty} \frac{5}{2} [\sin(at + bt) + \sin(at - bt)] e^{-st} dt \\ &= \frac{5}{2} \int_0^{\infty} \sin(at + bt) e^{-st} dt + \frac{5}{2} \int_0^{\infty} \sin(at - bt) e^{-st} dt \end{aligned}$$

$$F(s) = \frac{5}{2} \int_0^{\infty} [\sin(a+b)t] e^{-st} dt + \frac{5}{2} \int_0^{\infty} [\sin(a-b)t] e^{-st} dt \dots (1)$$

Step 3 of 5

Simplify the expression $\int_0^{\infty} [\sin(a+b)t] e^{-st} dt$.

It is known that,

$$\sin(\omega t) = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

Therefore,

$$\begin{aligned} \int_0^{\infty} (\sin(a+b)t) e^{-st} dt &= \int_0^{\infty} \left(\frac{e^{j(a+b)t} - e^{-j(a+b)t}}{2j} \right) e^{-st} dt \\ &= \frac{1}{2j} \left[\int_0^{\infty} e^{j(a+b)t} e^{-st} dt - \int_0^{\infty} e^{-j(a+b)t} e^{-st} dt \right] \\ &= \frac{1}{2j} \left[\int_0^{\infty} e^{-(s-j(a+b))t} dt - \int_0^{\infty} e^{-(s+j(a+b))t} dt \right] \\ &= \frac{1}{2j} \left[\left[\frac{e^{-(s-j(a+b))t}}{-(s-j(a+b))} \right]_0^{\infty} - \left[\frac{e^{-(s+j(a+b))t}}{-(s+j(a+b))} \right]_0^{\infty} \right] \end{aligned}$$

Further simplification as follows,

$$\begin{aligned} \int_0^{\infty} (\sin(a+b)t) e^{-st} dt &= \frac{1}{2j} \left[\left(\frac{e^{-(s-j(a+b))(\infty)}}{-(s-j(a+b))} - \frac{e^{-(s-j(a+b))(0)}}{-(s-j(a+b))} \right) - \left(\frac{e^{-(s+j(a+b))(\infty)}}{-(s+j(a+b))} - \frac{e^{-(s+j(a+b))(0)}}{-(s+j(a+b))} \right) \right] \\ &= \frac{1}{2j} \left[\left(\frac{0}{-(s-j(a+b))} - \frac{1}{-(s-j(a+b))} \right) - \left(\frac{0}{-(s+j(a+b))} - \frac{1}{-(s+j(a+b))} \right) \right] \\ &= \frac{1}{2j} \left[\frac{1}{(s-j(a+b))} - \frac{1}{(s+j(a+b))} \right] \\ &= \frac{1}{2j} \left[\frac{s+j(a+b) - s-j(a+b)}{(s-j(a+b))(s+j(a+b))} \right] \\ &= \frac{1}{2j} \left[\frac{2j(a+b)}{s^2 + (a+b)^2} \right] \\ &= \frac{(a+b)}{s^2 + (a+b)^2} \end{aligned}$$

Simplify the expression $\int_0^{\infty} [\sin(a-b)t] e^{-st} dt$.

It is known that,

$$\sin(\omega t) = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

Therefore,

$$\begin{aligned} \int_0^{\infty} (\sin(a-b)t) e^{-st} dt &= \int_0^{\infty} \left(\frac{e^{j(a-b)t} - e^{-j(a-b)t}}{2j} \right) e^{-st} dt \\ &= \frac{1}{2j} \left[\int_0^{\infty} e^{j(a-b)t} e^{-st} dt - \int_0^{\infty} e^{-j(a-b)t} e^{-st} dt \right] \\ &= \frac{1}{2j} \left[\int_0^{\infty} e^{-(s-j(a-b))t} dt - \int_0^{\infty} e^{-(s+j(a-b))t} dt \right] \\ &= \frac{1}{2j} \left[\left[\frac{e^{-(s-j(a-b))t}}{-(s-j(a-b))} \right]_0^{\infty} - \left[\frac{e^{-(s+j(a-b))t}}{-(s+j(a-b))} \right]_0^{\infty} \right] \end{aligned}$$

Further simplification as follows,

$$\begin{aligned} \int_0^{\infty} (\sin(a-b)t) e^{-st} dt &= \frac{1}{2j} \left[\left(\frac{e^{-(s-j(a-b))(\infty)}}{-(s-j(a-b))} - \frac{e^{-(s-j(a-b))(0)}}{-(s-j(a-b))} \right) - \left(\frac{e^{-(s+j(a-b))(\infty)}}{-(s+j(a-b))} - \frac{e^{-(s+j(a-b))(0)}}{-(s+j(a-b))} \right) \right] \\ &= \frac{1}{2j} \left[\left(\frac{0}{-(s-j(a-b))} - \frac{1}{-(s-j(a-b))} \right) - \left(\frac{0}{-(s+j(a-b))} - \frac{1}{-(s+j(a-b))} \right) \right] \\ &= \frac{1}{2j} \left[\frac{1}{(s-j(a-b))} - \frac{1}{(s+j(a-b))} \right] \\ &= \frac{1}{2j} \left[\frac{s+j(a-b) - s+j(a-b)}{(s-j(a-b))(s+j(a-b))} \right] \\ &= \frac{1}{2j} \left[\frac{2j(a-b)}{s^2 + (a-b)^2} \right] \\ &= \frac{(a-b)}{s^2 + (a-b)^2} \end{aligned}$$

Recall equation (1),

$$F(s) = \frac{5}{2} \int_0^{\infty} [\sin(a+b)t] e^{-st} dt + \frac{5}{2} \int_0^{\infty} [\sin(a-b)t] e^{-st} dt$$

Substitute $\frac{(a+b)}{s^2+(a+b)^2}$ for $\int_0^{\infty} (\sin(a+b)t) e^{-st} dt$ and $\frac{(a-b)}{s^2+(a-b)^2}$ for $\int_0^{\infty} (\sin(a-b)t) e^{-st} dt$ in the equation.

$$\begin{aligned} F(s) &= \frac{5}{2} \left(\frac{(a+b)}{s^2+(a+b)^2} \right) + \frac{5}{2} \left(\frac{(a-b)}{s^2+(a-b)^2} \right) \\ &= 2.5 \left(\frac{(a+b)}{s^2+(a+b)^2} \right) + 2.5 \left(\frac{(a-b)}{s^2+(a-b)^2} \right) \end{aligned}$$

Thus, the Laplace transform of $5 \sin(at) \cos(bt)$ is,

$$\boxed{2.5 \left(\frac{(a+b)}{s^2+(a+b)^2} \right) + 2.5 \left(\frac{(a-b)}{s^2+(a-b)^2} \right)}.$$